Multimedia Signal Processing - 2nd Module

July, 20 2017

1. **Question (6 pts)**
   Consider a source X emitting the following sequence of symbols:

   $$a\,b\,b\,b\,a\,b\,a\,a\,b\,b\,a\,b\,a\,c\,a\,c\,b\,b$$

   (a) Learn the probability mass function (p.m.f.) of the source from the observed data and compute the entropy $H(X)$ of the source (assuming a memoryless source).

   (b) Compute Huffman code in the scenario depicted in point (a)

   (c) Assume now that X is a source with memory, group symbols in blocks of size $N = 2$ and compute entropy $H_2(X)$.

   (d) Compute Huffman code in the scenario depicted in point (c)

   (e) Considering Huffman codes derived in points (b) and (d), write the codewords associated to the sequence $b\,b\,a\,b\,a\,c$. Which is their length in bits?

2. **Question (6 pts)**
   (a) Introduce the concept of perceptual audio coding (e.g., why it works, how it relates to the hearing system, when it can be applied successfully, different kinds of masking, etc.).

   (b) Describe the MPEG1 psychoacoustic model 1 for global masking threshold computation, possibly adding some sketches of spectral behavior.

3. **Question (6 pts)**
   (a) Illustrate the block diagram of a hybrid DCT/DPCM video codec (i.e., encoder and decoder). Clearly indicate which is the encoder output needed by the decoder to correctly reconstruct the video sequence.

   (b) Explain the concept of motion vectors, reporting some possible strategies to compute them and indicating why they are important in video coding.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Consider JPEG image coding scheme.

   - [T] [F] Either DCT or KLT can be used as block transform.
   - [T] [F] Large quantization steps are typically applied to low-frequency components.
   - [T] [F] Block size is fixed at 8x8 pixel by the standard.
   - [T] [F] Run-level coding is applied to DC components after quantization.

   (b) Consider a transform coding scheme.

   - [T] [F] Using unitary transforms, coding gain with respect to PCM is always greater than or equal to one.
   - [T] [F] DCT provides higher coding gain than KLT.
   - [T] [F] DFT cannot be used as it is a complex transform.

   (c) Consider two discrete memoryless sources X and Y emitting sequences of samples from a finite alphabet.

   - [T] [F] If alphabet of X contains more symbols than alphabet of Y, it is always true that $H(X) \geq H(Y)$ (where H means entropy).
   - [T] [F] The maximum entropy $H(X)$ of source X depends on the amount of symbols in its alphabet.
   - [T] [F] If X is a binary source, its entropy can be lower than $\log_2(2)$.

5. **Question (Matlab - 10pts)**

   Consider the mono audio track `gb.wav`.

   (a) Load the first 2 seconds of th audio track into variable `x`. Display the loaded waveform on screen.

   (b) Encode `x` using PCM considering four different uniform quantizers with rate 2, 4, 8, and 12 bits each. Adapt quantization levels to waveform dynamic range.

   (c) Draw two PCM rate distortion curves considering: i) SNR vs. quantizer bitrate; ii) SNR vs. entropy of the PCM encoded track. Compare and comments both curves.

   (d) Encode `x` using DPCM with memory one and considering four different uniform quantizers with rate 2, 4, 8, and 12 bits each. Encode the first sample using PCM, and approximately adapt quantization levels to signal residual dynamic range.

   (e) Draw two DPCM rate distortion curves considering: i) SNR vs. quantizer bitrate; ii) SNR vs. entropy of the DPCM encoded track. Compare and comment PCM and DPCM curves.